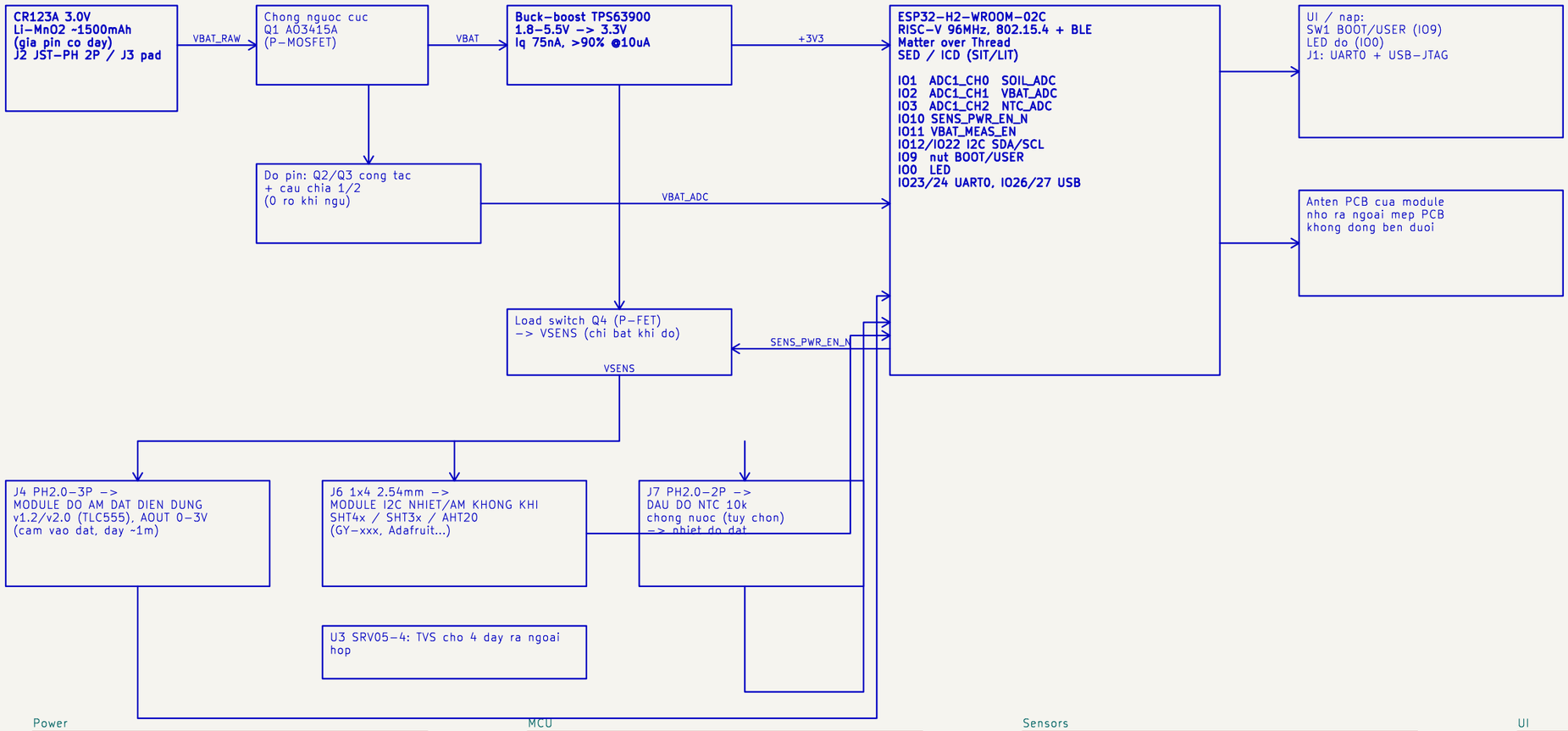
SO DO KHOI v2 — Bo mach ESP32-H2 + nguon pin, cam bien la module co san (Matter over Thread)



Power

Power: CR123A,
RPP,
TPS63900 3V3,
VBAT sense,
VSENS

File: power.kicad_sch

MCU

SOIL_ADC / NTC_ADC / I2C

MCU: ESP32-H2-WROOM-02C (Thread ICD)

File: mcu.kicad_sch

Sensors

Sensor connectors: soil module,
I2C module,
NTC probe

File: sensors.kicad_sch

UI

UI: button,
LED,
prog header,
test points

File: ui.kicad_sch

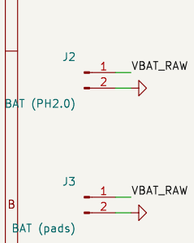
Root sheet: so do khoi + 4 sheet con (ket noi bang global label)
H2 Soil Sensor

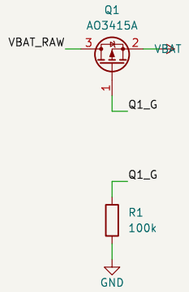
Sheet: /
File: h2-soil-sensor.kicad_sch

Title: H2 Soil Sensor — Matter over Thread (ESP32-H2, CR123A)

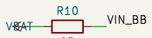
Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/5

1) PIN + CHONG NGUOC CUC
Pin CR123A (gia co day) noi vao J2 (JST-PH 2P) HOAC han
vao 2 pad J3. Q1 P-MOSFET: D->pin(+), S->tai, G->GND qua R1.
Lap nguoc pin: Vgs=0 -> khong co dong. R10 0R: cau do dong.

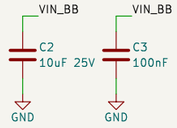


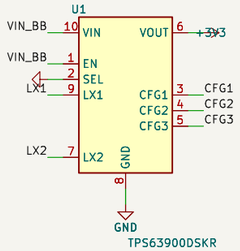


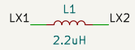




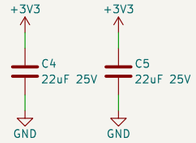
2) TPS63900 BUCK-BOOST -> 3.3V (Iq 75nA)
SEL=GND -> VO(1) theo CFG3=16.2k = 3.3V
CFG1=36.5k (VO(2)=3.3V), CFG2=0R: Iin khong gioi han
R CFG 1%/100ppm, khong dat tu tren chan CFG.
L1 2.2uH Isat>=2A; COUT 2x22uF; CIN 10uF.

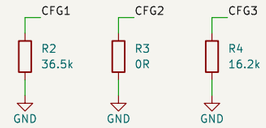


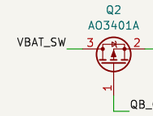


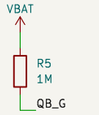


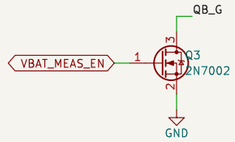
3) DO DIEN AP PIN (khong ro khi ngu)
VBAT_MEAS_EN=1 -> Q3 ON -> Q2 ON -> VBAT/2 -> IO2.
Khi ngu: R6 giu Q3 OFF, R5 giu Q2 OFF.

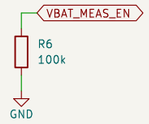


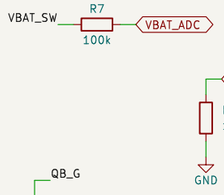




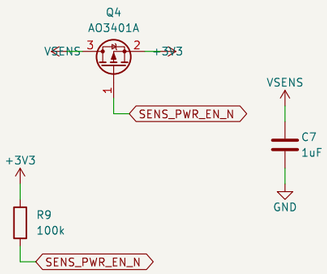








4) CONG TAC NGUON MODULE CAM BIEN VSENS
SENS_PWR_EN_N=0 -> Q4 ON: cap 3.3V cho module dat,
module I2C, NTC, pull-up I2C. R9 giu Q4 OFF khi
GPIO Hi-Z (sleep/reset). Module chi duoc cap nguon khi do.

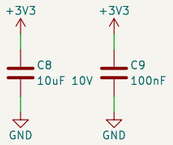


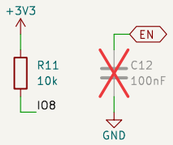
PWR_FLAG (cho ERC)

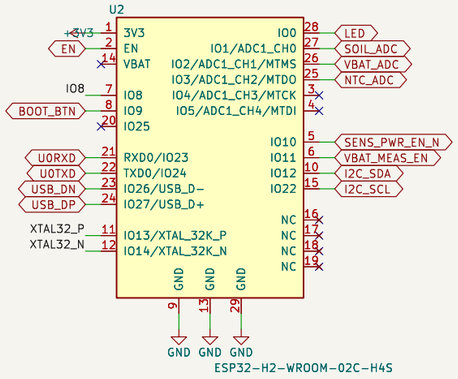


ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread
Sheet: /Power/
File: power.kicad_sch
Title: Power: CR123A, RPP, TPS63900 3V3, VBAT sense, VSENS
Size: A3 | Date: 2026-11-11 | **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/5

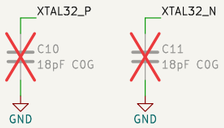
ESP32-H2-WROOM-02C (4MB: -H4S de co OTA). Vdd 3.0-3.6V.
Module da co RC tren EN (10k/1uF) -> C12 DNP.
Strapping: IO9=1 (R21, sheet UI) -> SPI boot; IO8=1 (R11).
IO10/IO11 la LP GPIO: giu muc khi light-sleep tat mien TOP.
ADC1: IO1=SOIL (module), IO2=VBAT, IO3=NTC. IO4/IO5/IO25 du phong.









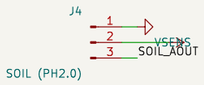


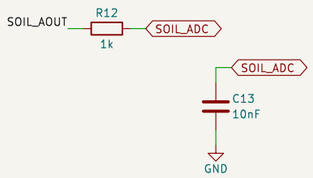
Anten PCB: module o mep tren, vung anten (6mm) nho ra ngoai PCB;
khong dong/GND/linh kien duoi va quanh anten; vo hop cach anten >=15mm.
Y1/C10/C11 (DNP): XTAL 32.768kHz tuy chon tren IO13/IO14.

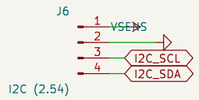
ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

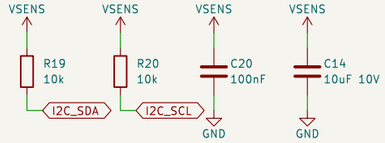
Sheet: /MCU/		
File: mcu.kicad_sch		
Title: MCU: ESP32-H2-WROOM-02C (Thread ICD)		
Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

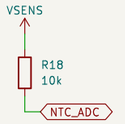
KET NOI MODULE CAM BIEN (tat ca nuoi tu VSENS, chi bat khi do)
J4: module do am dat dien dung v1.2/v2.0 (TLC555). day PH2.0-3P: 1=GND 2=VCC 3=AOUT.
    Chon module dung TLC555 (NE555 khong chay o 3.3V). AOUT 0-3V -> R12 -> IO1 (ATTEN 12dB).
J6: module I2C SHT4x/SHT3x/AHT20 (GY-xxx): 1=VIN 2=GND 3=SCL 4=SDA.
J7: dau do NTC 10k chong nuoc (tuy chon) -> nhiet do dat. U3: TVS cho day ra ngoai hop.

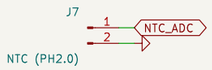


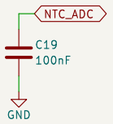


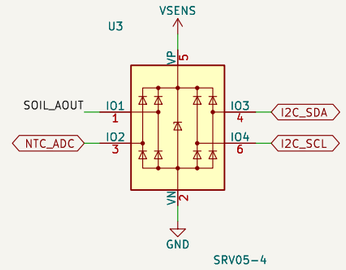












Pull-up I2C noi VSENS: firmware dat SDA/SCL Hi-Z (khong pull) truoc khi ngu.
Neu module I2C co LED nguon, nen go LED de giam dong khi do.

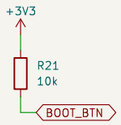
ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

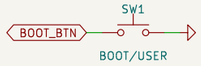
Sheet: /Sensors/
File: sensors.kicad_sch

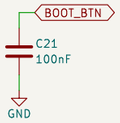
Title: Sensor connectors: soil module, I2C module, NTC probe

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

SW1 (IO9/BOOT): 1 lan = do & bao cao ngay / mo commissioning:
2 lan = hieu chuan KHO; 3 lan = hieu chuan UOT; giu >=10s = factory
reset. Giu SW1 khi cap nguon/reset = che do nap (download).

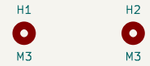


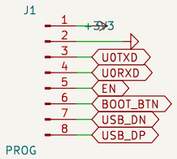












J1: 1=3V3 2=GND 3=TXD0 4=RXD0 5=EN 6=IO9 7=USB_D- 8=USB_D+
USB-Serial/JTAG: lap pin + noi D+/D-/GND (khong noi 3V3).
UART: thao pin, cap 3V3 tu adapter USB-UART 3.3V.

ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

Sheet: /UI/		
File: ui.kicad_sch		
Title: UI: button, LED, prog header, test points		
Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/5